Day – 7 Matplotlib and Seaborn

Matplotlib Tutorial

Matplotlib is an open-source library for creating static, animated and interactive visualizations in Python. Its object-oriented API enables the embedding of plots into applications developed with GUI toolkits such as Tkinter, Qt and GTK.

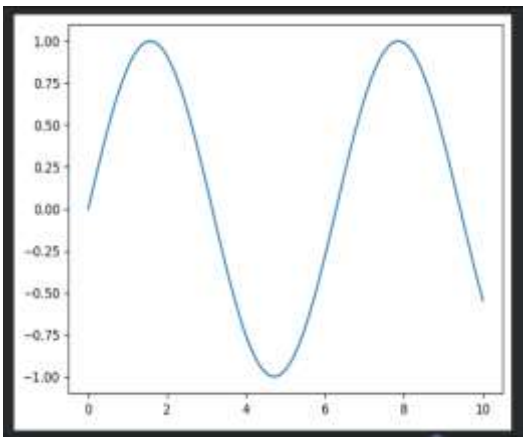
- It supports line plots, bar charts, histograms, scatter plots and 3D visualizations.
- Provides full command over figure elements, including axes, labels, legends and colors.
- Built to work with NumPy arrays and Pandas DataFrames.
- Compatible with multiple general-purpose GUI toolkits (Tkinter, Qt, GTK and wxPython).
- Capable of generating high-quality output in various hardcopy formats (PNG, PDF, SVG).

Pyplot

- Most of the Matplotlib utilities lies under the pyplot submodule, and are usually imported under the plt alias:
- import matplotlib.pyplot as plt
- Now the Pyplot package can be referred to as plt.

Example:

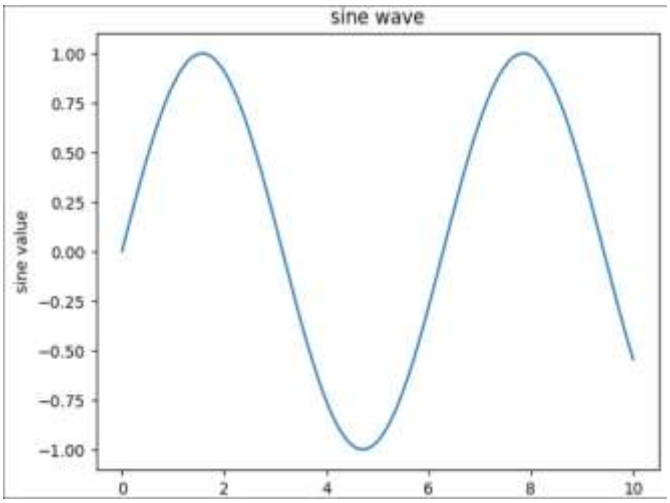
```
import matplotlib.pyplot as plt
import numpy as np
x=np.linspace(0,10,100)
y=np.sin(x)
z=np.cos(x)
print(x)
print(y)
plt.plot(x,y)
plt.show()
plt.plot(x,z)
plt.show()
```



Create Labels for a Plot:

With Pyplot, you can use the xlabel() and ylabel() functions to set a label for the x- and y-axis.

```
#adding title x-axis and y-axis labels
plt.plot(x,y)
plt.title('sine wave')
plt.xlabel('angle')
plt.ylabel('sine value')
plt.show()
```

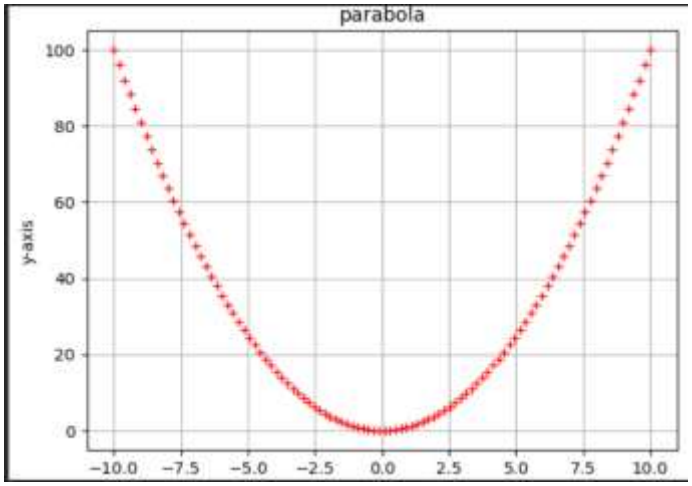


Add Grid Lines to a Plot:

With Pyplot, you can use the grid() function to add grid lines to the plot.

```
#Parabola
x=np.linspace(-10,10,100)
y=x**2
plt.plot(x,y)
```

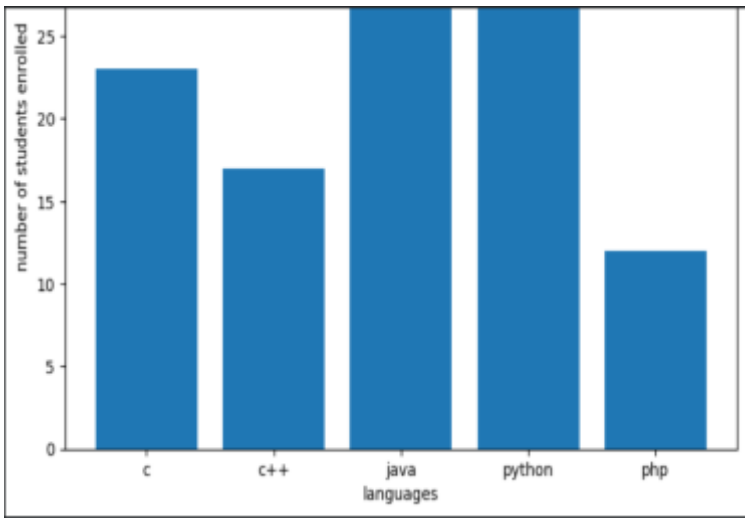
```
plt.plot(x,y,'r+')
plt.title('parabola')
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.grid(True)
```



Creating Bars:

With Pyplot, you can use the bar() function to draw bar graphs:

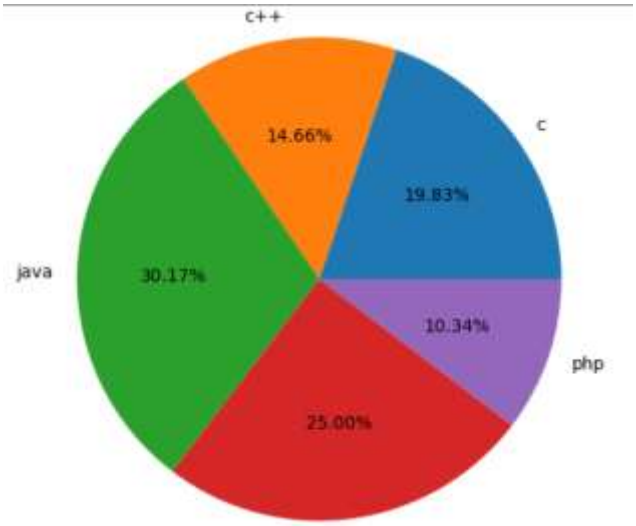
```
fig=plt.figure()
ax=fig.add_axes([0,0,1,1])
languages=['c','c++','java','python','php']
students=[23,17,35,29,12]
plt.xlabel('languages')
plt.ylabel('number of students enrolled')
ax.bar(languages,students)
plt.show()
```



Creating Pie Charts:

With Pyplot, you can use the pie() function to draw pie charts:

```
fig1=plt.figure()
ax1=fig1.add_axes([0,0,1,1])
languages=['c','c++','java','python','php']
students=[23,17,35,29,12]
ax1.pie(students,labels=languages,autopct='%1.2f%%')
plt.show()
```

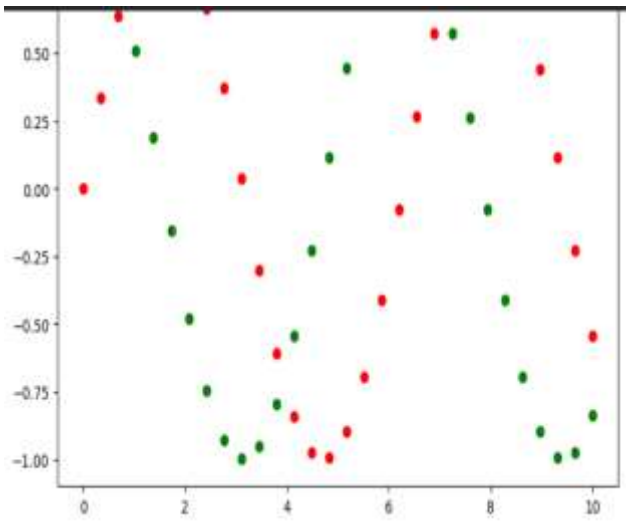


Creating Scatter Plots

With Pyplot, you can use the scatter() function to draw a scatter plot.

The scatter() function plots one dot for each observation. It needs two arrays of the same length, one for the values of the x-axis, and one for values on the y-axis:

```
x=np.linspace(0,10,30)
y=np.sin(x)
z=np.cos(x)
fig2=plt.figure()
ax2=fig2.add_axes([0,0,1,1])
ax2.scatter(x,y,color='r')
ax2.scatter(x,z,color='g')
plt.show()
```



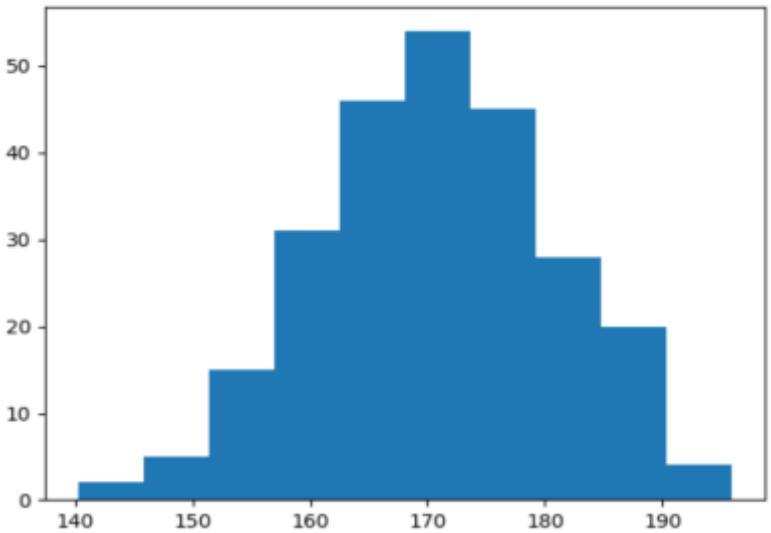
Histogram

A histogram is a graph showing frequency distributions.In Matplotlib, we use the hist() function to create histograms.The hist() function will use an array of numbers to create a histogram, the array is sent into the function as an argument

```
import matplotlib.pyplot as plt
import numpy as np

x = np.random.normal(170, 10, 250)

plt.hist(x)
plt.show()
```



Seaborn

Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.

- Provides a layer of abstraction hence simpler to use
- better aesthetics
- more graphs included

Main Components of Seaborn:

Types of Functions 1. Figure Level 2. Axis Level

Main Classification - Relational Plot - Distribution Plot - Categorical Plot - Regression Plot - Matrix Plot – Multiplots

Example:

1. Relational Plot

Generally used for - see the statistical relation between 2 or more variables. - Bivariate Analysis

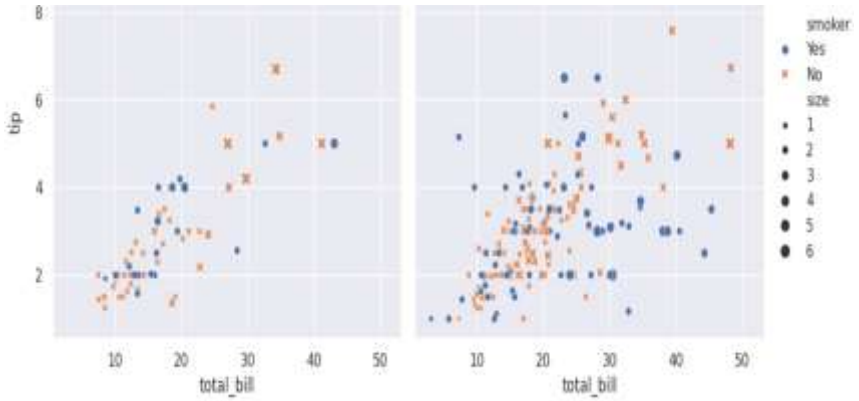
Figure level - relplot()

Axis level - 1. scatterplot() 2. lineplot()

```
#total bill vs tip dataset
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
tips=sns.load_dataset('tips')
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

```
#visualize the data
sns.relplot(data=tips,x='total_bill',y='tip',col='time',hue='smoker',style='smoker',size='size')
sns.set_theme(style='darkgrid')
```



2. Distribution Plots:

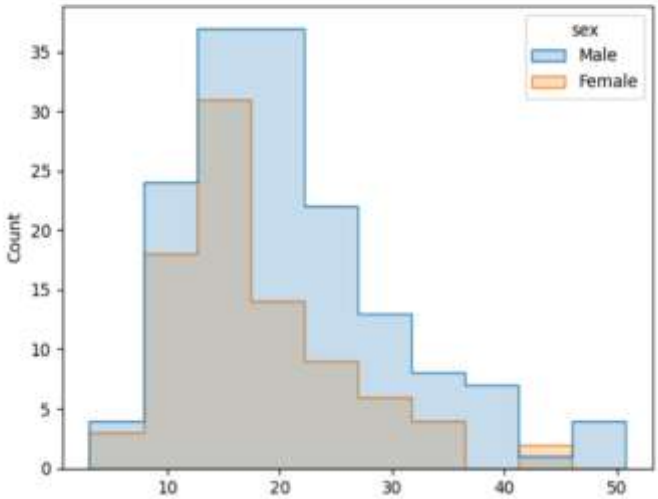
Generally used for - Univariate Analysis - used to find out the distribution of data - shows the range of the data - central tendency - used to check whether there are any outliers or the data bimodal?

Figure level - displot()

Axis level - 1. histplot() 2. kdeplot() 3. rugplot()

```
sns.histplot(data=tips, x='total_bill', bins=10, hue='sex', element='step')
```

```
<Axes: xlabel='total_bill', ylabel='Count'>
```

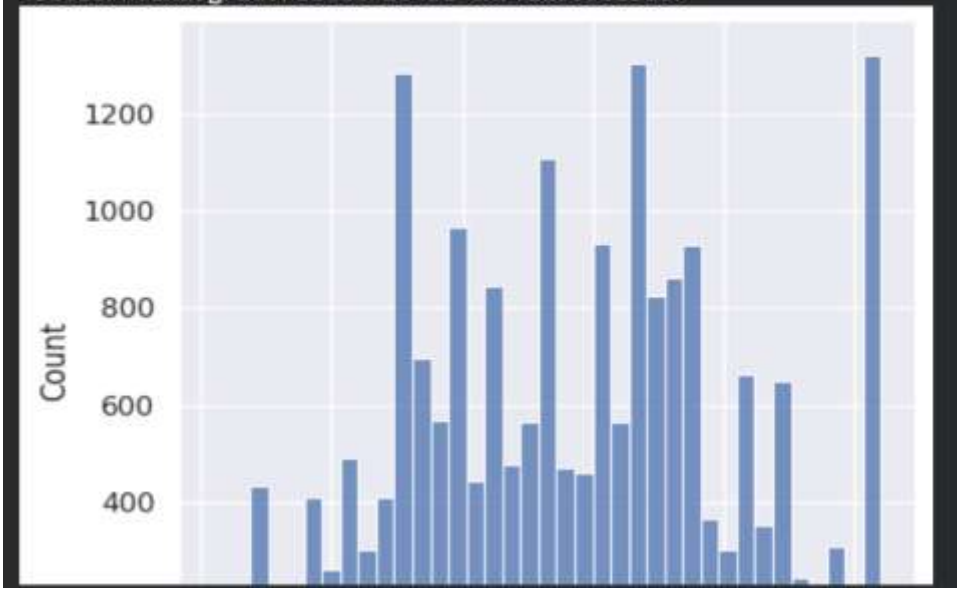


```
# house price dataset
from sklearn.datasets import fetch_california_housing
housing = fetch_california_housing()
type(housing)
print(housing)
# You can access it just like you tried with Boston:
print(housing.data.shape)
print(housing.target)
cali_df=pd.DataFrame(housing.data,columns=housing.feature_names)
cali_df.head()
```

```
{'data': array([[   8.3252    ,   41.        ,    6.98412698, ...,    2.55555556,
          37.88      , -122.23      ],
       [   8.3014    ,   21.        ,    6.23813708, ...,    2.10984183,
          37.86      , -122.22      ],
       [   7.2574    ,   52.        ,    8.28813559, ...,    2.80225989,
          37.85      , -122.24      ],
       ...,
       [   1.7       ,   17.        ,    5.20554273, ...,    2.3256351 ,
          39.43      , -121.22      ],
```

-
```
sns.displot(data=cali_df,x='HouseAge')
```

```
<seaborn.axisgrid.FacetGrid at 0x7f129056a5d0>
```



Heatmap

Heatmap is used to visualize matrix-like data using colors. It is commonly used for correlation matrices.

```
#constructing a Heat map
correlation=cali_df.corr()

plt.figure(figsize=(10,10))
sns.heatmap(correlation,cbar=True,square=True,fmt='.1f',annot=True,annot_kws={'size':8},cmap='Blues')

plt.show()
```


Population
AveOccup
Latitude
Longitude
MedInc
HouseAge
AveRooms
AveBedrms
Population
AveOccup
Latitude
Longitude
0.00
−0.25
−0.50
−0.75